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MILITARY SPECIFICATION

DRIER-MIXERS, BITUMINOUS-CONCRETE

MATERIALS, WHEEL MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers trailer-mounted, engine-driven drier-mixers that produce high-grade hot bituminous mixes.

1.2 Classification. The drier-mixers shall be of the following types, as specified (see 6.2):

Type I - Separate drier drum and pugmill.
Type II - Combination drying and mixing chamber.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-S-781	- Strapping, Steel, Flat and Seals.
TT-P-664	- Primer Coating, Synethetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, 30-Volt, Heavy Duty, Single-Conductor and Multiconductor.
MIL-C-13486/12	- Cable, Special Purpose, Electrical, Low Tension, Heavy Duty Multiple Conductor: Unshielded.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative and Break-In.

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MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-B-52919	- Belts, V. Industrial Drive, Narrow Type
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35643	- Gasket: Cap, Fuel Tank.
MS35644	- Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank-Tactical Vehicle, Fording.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular 28-Volt, Waterproof.

(Copies of specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specifications to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the US Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 Photochemically Reactive Solvents

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA Recommanded Practices.
TTMA Technical Bulletins

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., Suite 203, 2430 Pennsylvania Ave., N.W. Washington, D.C. 220037.)

NATIONAL MOTOR FREIGHT TRAFFIC CLASSIFICATION

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Department, 1616 P Street, NW, Washington, D.C. 20036.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to Tire and Rim Association, Inc., Comand Building, 34 North Hawkins Avenue, Akron, OH 44313.)

AMERICAN ASSOCIATION OF STATE HIGHWAY OFFICIALS (AASHO)

Designation T58 - Asphalt Extraction Test.

(Application for copies should be addressed to the American Association of State Highway Officials, National Press Building, Washington, D.C. 20004).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

B29.1 - Precision Power Transmission Roller Chains, Attachments and sprockets.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

ASPHALT INSTITUTE, INC.

Asphalt Plant Manual, Series 3.

(Application for copies should be addressed to the Asphalt Institute, Inc., Asphalt Institute Building, Campus, University of Maryland, College Park, MD 20740.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The drier-mixer shall consist essentially of an engine, charging hopper, bitumen metering pump, bitumen storage tank, burner heating system, and pugmill installed on a pneumatic-tired, cart-type trailer. Necessary controls, drives, and accessories shall be provided. Controls shall be located so as to provide for continuous operation by one man. Except as specified herein, the drier-mixer automotive type components, assemblies and accessories shall be commercial automotive products and shall comply with all Federal Motor Vehicle Safety Standards applicable to nonmilitary, nonexport motor vehicle products on the date of manufacture.

3.1.1 Weights and dimensions. The gross vehicle weight (GVW) of each drier-mixer shall include the weight of all machinery, equipment, accessories, and attachments; with all tanks full; and with complete complement of fuel, coolant, and lubricants. Component and vehicular ratings shall not be raised to conform to the requirements of this specification. The physical characteristics of each unit, prepared for transport, shall be as specified in table I, with the drier-mixer on level ground and with the inflation pressure of the tires as recommended in the Tire and Rim Association Yearbook for the imposed GVW loading.

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TABLE I. Weights and dimensions

Characteristic	Requirement
GVW	11,000 lbs maximum
Vertical towbar load, range	5 to 12 percent of GVW
Overall length	260 inches, maximum
Overall width	96 inches, maximum
Overall height (shipping)	102 inches, maximum
Ground clearance, exclusive of suspension	14 inches, minimum
Towbar eye height range (SAE J849)	27 thru 33 inches

3.1.2 Performance. The drier-mixer shall dry and mix sand, gravel or crushed stone with an 85 to 100 penetration grade asphaltic material. Operating at continuous production, the minimun output capacity of the drier-mixer for hot bituminous mixes when tested as specified (see 4.5.2.7), shall be as shown in table II. For other bituminous mixes, the capacity of the drier-mixer shall be not less than 10 tons per hour (see 4.5.2.7.2). Prior to mixing with bitumen, the residual surface moisture of the aggregate shall be not more than 1 percent by weight (hot mix). The temperature of the aggregate at the end of the drying cycle or at drier discharge shall be not more than 325° F, and not less than 275° F. (hot mix).

TABLE II. Hot bituminous mix output.

Initital surface mositure content, Percent	Continuous output capacity, minimum tons/hr
3	8
4	7
5	6
6	5
7	4
8	3

3.1.3 Compatibility. The drier-mixer shall be compatible for towing behind a 5 ton, military design dump truck or equal and shall operate in conjunction with the dump truck (in towing position) so that the aggregate can be shoveled into the charging hopper from the dump truck. The Government will loan the contractor a dump truck (see 6.3).

3.2 First article (preproduction model). Unless otherwise specified (see 6.2), the contractor shall furnish one or more drier-mixers for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce drier-mixers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the

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contractor subject to surveillance and approval by the Government (see 6.4). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Production drier-mixers. All drier-mixers furnished under this specification shall be physically and mechanically identical to the first article drier-mixer approved by the Government (see 6.4).

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8). Materials and special treatments utilized shall provide resistance to rust and corrosion.

3.5 Safety and human factors.

3.5.1 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature that are of such nature or are so located as to be, or become, a hazard to operating or maintenance personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. All exhausts shall be directed so as not to endanger personnel or obstruct the view of the operator. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine, exhaust, aggregate drier, or electrical equipment. Electrical equipment shall be insulated, fully enclosed, or guarded. Nonfunctional sharp edges, projecting points and protruding length more than two full threads of fastening devices shall be avoided.

3.5.2 Human factors. The characteristics of the drier-mixer shall provide for operation by personnel ranging from the small man clothed, through the large man clothed in accordance with SAE J833.

3.5.3. Noise level.

3.5.3.1 Exterior noise. The exterior sound level of the drier mixer shall not exceed 88dB(A) when measured in accordance with SAE J88.

3.5.3.2 Personnel noise. The sound level at the operator's position shall not exceed 85dB(A) as specified for Category D in MIL-STD-1474. It shall be tested in accordance with 4.5.2.11.2.

3.6 Maintenance

3.6.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear

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cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using special tools furnished with the drier-mixer and common tools. After completion of any maintenance operation, the drier-mixer shall operate as specified herein.

3.6.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.7 Trailer. The drier-mixer shall be mounted on a pneumatic-tired, 2-wheel cart-type trailer chassis that will provide on-highway and secondary road mobility specified as follows when towed by the dump truck specified in 3.1.3.

3.7.1 Mobility. The drier-mixer shall perform as specified when: (1) equipped with all attachments, equipment, and accessories stowed for travel and with all tanks filled to capacity; (2) towed over highways, and secondary roads.

3.7.1.1 Highway travel. The drier-mixer shall travel over smooth, hard surfaced, primary highways at sustained speeds up to 55 miles per hour (mph) without failure or permanent deformation.

3.7.1.2 Secondary road travel. The drier-mixer shall travel over gravel and improved roads at speeds up to 25 mph without failure or permanent deformation.

3.7.2 Trailing and turning ability. The drier-mixer shall trail the towing vehicle without bottoming, cramping, weaving, or sidesway. The drier-mixer shall follow the truck without exceeding the tracking deviation, as specified in DOT Federal Motor Carrier Safety Regulations, 393.70, when towed over a paved surface. Jackknife turns of at least 60 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck and drier-mixer.

3.7.3 Frame. The frame shall be of sufficient strength and rigidity to support and connect all components, maintain alinement, and distribute the stress of the suspension mounts in order to meet the transport and operating conditions specified herein.

3.7.4 Suspension. The drier-mixer shall be equipped with a spring, single axle suspension system, tailored to the drier-mixer. The suspension manufacturer's rated capacity shall be equal to, or greater than the load imposed on each member with the drier-mixer loaded to maximum GVW. When tested as specified in 4.5.2.2, the suspension shall provide a nondamaging ride on graveled-surfaced secondary roads and a stable ride on paved highways. The installation of the suspension shall be in correct alinement with the frame and with the axle so that the two diagonal measurements from the center of the lunette to the outer center of each spindle on the axle do not differ by more than 1/16-inch for every 10 feet or fraction thereof. Elastomer cushioned bumper

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stops or blocks shall be furnished to prevent axle-to-frame contact and limit suspension articulation at full compression (jounce) and rebound. Shock absorbers shall be furnished to control jounce and rebound.

3.7.5 Axle assembly. The axle shall be rated by the axle manufacturer at a continuous load capacity rating of not less than the drier-mixer weight when loaded to maximum GVW. The spindles shall be alloy steel hardened and ground and shall be located in the correct alinement position with the axle tube. The spindles shall be equipped with the assemblies of hubs, bearings and seals as specified herein. Hub assemblies shall be equipped with commercial tapered roller bearings having a standard inside-to-inside bearing spacing. Cups, bearings, and cones of all bearings shall be in matched sets from the same bearing manufacturer. The axle hubs shall be of the 6-stud, 8-3/4-inch bolt circle type, conforming to SAE J694.

3.7.6 Wheels, rims, and tires. The drier-mixer shall be equipped with single demountable disc wheel and tire assemblies. The disc wheels shall be of the heavy duty type with T&RA rim. The rim and wheel disc shall be constructed by the same wheel manufacturer and shall be in accordance with the wheel manufacturer's recommendations for the loads and service specified herein. Rims, tires, inner tubes, and flaps (if furnished), valve cores, and caps shall be in accordance with the T&RA Yearbook and supplementary service load imposed on each tire, measured on level ground under loaded travel conditions, shall not exceed the recommended rated loads and inflation pressures of the tires for 55 mph highway speed for continuous highway operation. Valve shall be of the C-4, heat-resistant types.

3.7.6.1 Balancing. Tires, wheels, hubs, and brake drums shall be in balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified herein.

3.7.7 Fenders and mudguards. The drier-mixer shall be equipped with two fenders and two rear splash guards. The fenders and splash guards shall restrict water and other road surface substances thrown from the drier-mixer wheels. The fenders shall include inner splash shields. The fenders shall support a weight of not less than 200 pounds imposed on any 1-square-foot area at any one time without damage or permanent deformation. The outer edges of the fenders shall be rolled to reduce side road splash. The splash guards shall be of standard commercial grade and shall be of a size not less than 12 by 18 inches, and shall include antisail characteristics.

3.7.8 Intervehicular connections. The drier-mixer shall be furnished with intervehicular electrical harness and two airbrake lines for connecting the truck's electrical and airbrake systems to the drier-mixer systems. All intervehicular connections shall be installed on the drier-mixer so that they extend forward between 36 and 42 inches toward the towing vehicle from the towbar eye. Retaining devices shall be installed to hold the electrical harness plugs to the towbar with the harness doubled back on the drier-mixer. Two dummy couplings shall be installed to hold the airbrake line ends with the lines

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doubled back. When a 12 and 24 volt vehicular electrical system is specified, intervehicular harness compatible with each system shall be provided as specified herein.

3.7.9 Towbar components. A towbar eye conforming to SAE J847 shall be furnished and installed on the towbar within 1/8-inch tolerance of the longitudinal centerline of the drier-mixer. The towbar shall be within the height range specified in table I. The lunette (towbar eye) shall be inserted in a support that is welded into the towbar. The lunette support shall prevent lunette rotations. Two safety chains, equipped with safety type grabhooks, shall be furnished and shall conform to SAE J697. The forward length of each chain hook shall be as specified in 3.7.8. The attachment end of the safety chains shall be perimeter welded to the towbar. The towbar shall be constructed to embody structural strength to meet or exceed the following requirements: Support and transport the weight of the drier-mixer at GVW (see table I); transmit and distribute the imposed load to the suspension system and eye and resist bending and twisting without permanent deformation or part failure when subjected to the performance requirements specified herein.

3.7.10 Landing gear. A single-leg landing gear, shall be installed on the towbar of the drier-mixer. The landing gear shall be vertically adjustable for not less than 12 inches. The landing gear shall either retract or be hinged and fold to provide not less than the ground clearance of the first trailer crossmember. The landing gear shall be arranged for positive locking into park and travel positions. A folding-type crank handle for operating the vertical adjustment mechanism shall be provided. The tangential driver's effort at the handgrip to raise the maximum GVW load shall not exceed 30 pounds. The mechanism shall be protected from exposure to foreign matter which would impair functional efficiency. The landing gear shall incorporate a self-aligning pad which will provide a ground bearing load not exceeding 20 pounds per square inch when the maximum towbar weight in table I is imposed. The edges of the pad shall be turned up.

3.7.11 Rear support leg. An adjustable support leg shall be furnished and shall be located at the left or right rear corner of the drier-mixer. The support leg shall be in accordance with the requirements specified for the landing gear in 3.7.10.

3.7.12 Brake system. The drier-mixer shall be equipped with an air or air activated brake system. The system shall conform to the DOT Motor Carrier Safety Regulations and, the DOT Motor Vehicle Safety Standards. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from ground obstruction, debris, and water and to obtain maximum safety, efficiency, and ease of maintenance. The brake system shall include service and emergency gladhands, relay emergency valve, straight air cylinders or chambers, foundation brakes on each wheel, and all other components and plumbing required for a complete matched brake system. The gladhands shall conform to SAE J318. The gladhands shall be colored coded yellow for service and red for emergency. The gladhands shall have a built in filter screen and rubberized dust seal. The dust seal shall not cause

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interference when the gladhand is coupled to towing vehicle gladhand. The gladhands shall be installed on the front of the trailer with the service on the curb side and emergency on the road side. In addition, brake line identification tags shall be furnished.

3.7.13 Chock blocks. Two chock blocks conforming to figure 1 shall be provided. The captive chain shall be hooked on the wire cradle's lower loop with excess chain wound in a figure eight manner around the upper ears, and when uncoiled from the cradle, shall permit the clocks to be placed before and behind the drier-mixer tires. Chock-block cradles conforming to figure 2 shall be provided for storing the chocks.

3.7.14 Vehicular electrical system. The drier-mixer shall be equipped with a waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following (in order of precedence):

(a) DOT Federal Motor Vehicle Safety Standards (FMV SS)
(b) DOT Federal Motor Carrier Safety Regulations (FMCSR)
(c) SAE Handbook, Electrical Equipment and Lighting Section.
(d) TTMA Recommended Practices (RP No. 6, RP No. 9, RP No. 14, RP No. 40, and Technical Bulletin No. 65.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The electrical components shall be in accordance with table III. Item quantities that are not specified shall be furnished as required (AR) for each drier-mixer.

TABLE III. Vehicular electrical system

	Requirements	Quantity Per Heater
7 Conductor Plug and Intervehicular Harness Assembly	SAE J560 and TTMA RP#40, 7-wire type with plated metal connectors, splash proof plug cover. SAE J1067 seven conductor jacketed cable fastened to connection box.	1 ea
Connection box	Splash proof type with stud and nut terminals.	1 ea min
Circuit breakers	SAE J553 protecting all live circuits.	AR
Wiring	Heavy duty type. Wire size shall conform to SAE J555.	AR
Wire identification	SAE J560 or J821.	AR

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TABLE III. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per Heater
Wiring harness	Polyethylene electrical conduit, each end weather proofed with silicone sealant or with one-half over-lapping turns of plastic electrical tape.	AR
Grommets	Provided where needed along the wire routing to prevent chafing.	AR
Supports	Elastomer cushioned support straps or elastomer ties with mounts; wiring support spacing 36 in. max. span, prevent whipping, chafing, and fatigue damage.	AR
Wiring splices	Mechanical splicing devices; interior waterproofed with silicone lube or compounds.	AR
Connectors and terminal	Waterproof types; blade type SAE J858, pin type SAE J928, and spade SAE J561. Blade and pin types fully enclosed and lubricated, prior to assembly with silicone lubricant; spade type equipped with heat shrinkable sleeves and coated with electrical varnish after assembly; all connectors and terminals plated or solder-coated to retard corrosion.	AR
Connector holder	Plastic or plastic-insulated metal spring clips; fastened to vehicle; furnished to anchor wire-to-wire connectors.	1 ea per connector
Running lamps	4 sidemarker, 4 clearance, 3 rear identification; per TTMA RP-108/1 or RP-108/3. Waterproof and shock resistant types.	11 ea min
Combination lamps	2 Turn and Tail outboard and 2 Stop and Tail inboard; Per TTMA RP No 14 and SAE J759; double contact; 3 wire type including ground wire; waterproof and shock resistant type.	4 ea min

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TABLE III. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per Heater
Lamp lenses and bulbs	Heavy duty or fleet service type bulbs, bulbs mounted in vibration-protected sockets with insulators and stainless steel or rubber bulb tension devices.	AR
Reflectors	2 on each corner, round type, securely fastened by bolts, screws or rivets.	8 ea min
Wiring diagram	Plate furnished on trailer per TTMA Bulletin #65.	1 ea

3.7.14.1 Conversion to 24 volt electrical system. Unless otherwise specified (see 6.2), the drier-mixer trailer shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact plug, the trailer shall be equipped with the following military intervehicle components:

(a) Military 12-contact plug - MS75020-1.
(b) Military intervehicular cable - MIL-C-13486/12 or SAE J1067 seven conductor jacketed cable.

The military plug and cable or SAE cable shall be the same length as the TTMA plug and cable and both shall be labeled with identification plates for 24-volts and 12- volts respectively, negative ground. The military 24-volt intervehicular cable shall be routed into a splashproof stud and nut type connection box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with table IIIA.

TABLE IIIA. Military-commercial connections.

Function	Mil. Plug Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Blackout marker	A	No.	-
Ser. left turn & stop	B	Connect to -	YEL (3)
Blackout marker	C	No.	-
Ground	D	Connect to -	WHT (1)
Ser. tail & marker	E	Connect to -	BLK (2)
		& Connect to -	BRN (6)
Blackout stop	F	No.	-
Blackout marker	H	No.	-

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TABLE IIIA. Military-commercial connections. (Continued)

Function	Mil. Plug Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Ser. right turn & stop	J	Connect to -	GRN (5)
Aux. power (hot)	K	No.	BLU (7)
Ground	L	Ground to Frame	-
Electric brake	M	No.	-
Electric brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the trailer's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the trailer's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.8 Engine. The drier-mixer be powered by a gasoline engine or diesel engine as specified (see 6.2) and shall be the same engine the contactor normally furnishes in his commercial drier-mixer. The engine shall be equipped with a 12 volt electric cranking system and a 25 amp battery charging system. Instruments and all other accessories shall be as specified herein.

3.8.1 Engine fuel tank. A fuel tank conforming to the contractor's commercial standard, except as specified herein shall be provided. The tank capacity shall be sufficient for 8 hours of operation at the maximum fuel rating of the engine. The tank shall be provided with fuel-tank cap gasket, filler neck, and filler-neck cap conforming to MS35643, MS35644, and MS35645, respectively. The tank and all lines shall be located or positioned to eliminate fire hazard caused by spilling, overflow, or drainage of fuel. The tank shall be stenciled: "FILL WITH GASOLINE ONLY" for gasoline powered drier-mixer or "FILL WITH DIESEL FUEL ONLY" for diesel powered drier-mixers. On diesel powered drier-mixers the burner fuel oil tank (see 3.10.6) may also be used as the engine fuel tank when the burner fuel oil tank meets the requirements specified herein.

3.8.2 Instrument panel. The instrument panel shall contain the following gages:

(a) Fuel level gage.
(b) Ammeter.
(c) Lubricating oil pressure gage.
(d) Coolant temperature gage (as applicable).
(e) Hourmeter-tachometer (combination unit).

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3.8.3 Storage batteries. Unless otherwise specified (see 6.2), storage batteries conforming to MS35000, shall be furnished. When gasoline engines are provided the batteries shall be type 2HN or 6TN. When diesel engines are provided the batteries shall be type 6TN. The batteries shall be shipped charged and dry.

3.8.3.1 Battery container. Batteries shall be contained and restrained in a corrosion-resisting and acid-resisting box or frame providing for gas venting at or near the top of the container, drainage, rigid battery positioning, and accessibility to and observation of the battery cells and terminals. Restraint of the batteries shall be provided by acid resistant nonconductive blocks or by holddown frames bearing on the battery cases. Adequate drainage shall be provided for water or electrolyte spilled within the container. Drainage shall be accomplished without draining over other parts or components. The battery container shall be constructed and located so that the storage batteries are accessible without removal of the batteries or disconnecting of the leads for examination of terminals and electrolyte level and for liquid replenishment. The battery terminals and clamps shall be at least 1 inch away from any uninsulated electrical conductors, except the battery cables and battery. With the container cover removed, clearance shall be maintained to permit connection and disconnection of the battery cables without danger of shorting from the battery to other components. Batteries shall be removable from the container without shorting the battery terminals against other components.

3.8.3.2 Battery cables and terminals. Battery cables shall conform to MIL-C-13486, type I, class A, numbers 00, 0 or 2 AWG cable size designation. Battery cables shall not be spliced. Clamp type, noncorrosive, heavy-duty terminals on the ends of cables connected to the battery shall be provided to facilitate disconnection and battery removal. Rubber terminal shields shall be provided and installed.

3.8.4 Engine fuel. The gasoline engine, when furnished, shall operate on unleaded gasoline conforming to VV-G-1690. The diesel engine, when furnished, shall operate on fuel oil conforming to VV-F-800, grade DF2.

3.8.5 Engine lubrication. The engine lubricating oil shall be in accordance with MIL-L-2104.

3.8.5.1 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.8.6 Clutch. A hand-operated, over-center, engagement-type clutch shall be furnished. The clutch shall have a load capacity rating of not less than the maximum engine torque.

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3.9 Bitumen-storage tank. A built-in storage tank shall be provided as an integral part of the drier-mixer and shall have a minimum capacity of 200 gallons. A heating system shall be provided to heat the contents of the storage tank. The heating system shall have a heating capacity that will raise the temperature of the contents of the storage tank from 200°F to 300°F in less than 60 minutes when the storage tank is filled with asphalt cement, grade 85 to 100 penetration. Provisions shall be made for circulating the asphalt cement while it is being heated. The bitumen tank shall be provided with a top and bottom discharge line. The top discharge line shall be located between 2 inches to 4 inches above the top of the heating flues. The bottom discharge line shall be located between the bottom of the heating flues and the bottom of the bitumen tank. The discharge lines shall be provided with the required valving to control the material flow. A thermometer shall be installed in the tank and shall indicate the average temperature of the bitumen by means of a temperature indicator. The temperature indicator shall be mounted on the drier-mixer in a position so that the readings are discernible to the operator standing on the ground in his normal operating position. A filler opening not less than 6 inches in diameter with a removable strainer and a filler cap shall be provided. The filler cap shall be secured to the strainer with a chain and swivel. Provisions shall be made for holding a 55-gallon drum of bitumen while the contents of the drum is being emptied into the storage tanks. All tank openings shall be covered and the covers shall prevent entry of rain and dust. The covers shall be equipped with quick-disconnect fasteners that will keep the cover closed. The tank shall be provided with an air vent and drain valve. The drain valve shall provide complete drainage of the tank. Drainage shall be to the ground when the drier-mixer is in its normal operating position. Integral pipes may be used to convey the bitumen from the drain outlet to the ground.

3.10 Heating system. The drier-mixer heating system shall provide the heat required to heat and dry aggregate and heat asphalt cement at a rate sufficient to meet the requirements specified herein. The heating system(s) shall consist essentially of burners, blower(s), combustion chamber(s), fuel pump, fuel tanks and exhaust stack(s).

3.10.1 Burners. The burners shall burn fuel oil conforming to VV-F-800, grade DF2, at a rate sufficient to meet the requirements specified herein. A control valve shall be installed in the fuel supply line to each burner at a location accessible to the operator. The control valve shall provide an infinitely variable range of adjustment from closed to the fully open position and shall maintain a steady flow of fuel at any setting.

3.10.2 Pressure gage. An oil pressure gage shall be installed in each burner fuel supply line. The gage shall be located in a position so that the readings are discernible to the operator standing on the ground in his normal operating position. The pressure gage shall have a metal case and a glass-covered indicating dial with a range of zero to a reading of not less than 125 percent greater than the normal operating pressure. The dial shall be not less than 3 inches in diameter and shall be graduated in increments of not more than 5 psi.

3.10.3 Blower. A centrifugal-type blower shall be provided and shall supply air for burner(s) fuel oil combustion. The blower shall have a capacity to

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provide a sufficient quantity of air to the burners to assure efficient operation of the burners at maximum burner capacity. Means for adjusting the air pressure shall be provided and shall provide a variable range of adjustments for maintaining a steady flow of air to each burner. The blower drive shall be in accordance with the requirements specified in 3.14.

3.10.4 Combustion chamber. The combustion chamber shall be fabricated of steel plate with a replaceable refractory lining or of heat-resistant airflow design without replaceable refractory lining.

3.10.5 Fuel pump. The fuel pump shall be of the rotary type. Capacity of the pump shall be not less than 125 percent of the actual burner requirements. A valve shall be provided to return excess fuel oil to the tank. The suction side of the pump shall be equipped with a strainer.

3.10.6 Fuel-oil tank. A fuel-oil tank conforming to the contractor's commercial standard, except as specified herein, shall be provided. The tank shall have a capacity of not less than 55 gallons and shall be made completely of a material which will not be corroded by water or fuel oil. The tank shall be provided with a fuel tank cap gasket, filler neck, and filler-neck cap conforming to MS35643, MS35644, and MS35645, respectively. The tank shall be provided with an air vent, a drain, a fuel-level indicator, and a fuel-line valve. The fuel tank outlet shall be located not less than 1/2-inch and not more than 1 inch above the bottom of the tank. The outlet fuel-line valve shall be located not more than 6 inches from the fuel tank. The tank shall be mounted to give protection from external damage and shall not present a fire hazard. All fuel lines shall be secured to prevent vibration and chafing, and shall have no loops or dips that might permit the accumulation of water. The tank shall be marked: "FILL WITH DIESEL FUEL ONLY".

3.10.7 Exhaust stacks. Exhaust stacks shall be provided for venting the bitumen storage tank heating system and aggregate drying chamber. The stack height shall be not less than 9 feet above ground level when the drier-mixer is in its normal operating position. The stack height shall be reduced for travel by hinging the stack or providing removable sections equipped with quick-release fasteners.

3.11 Portable burner. A portable generating-torch-type burner which burns fuel conforming to VV-F-800, grade DF2, shall be furnished. The burner shall be used for heating the pump and asphalt piping. The burner at maximum firing rate shall burn not less than 0.5 gallon of fuel per hour and shall have a fuel tank capacity of not less than 2 gallons or be connected to the fuel pump (see 3.10.5). The tank when supplied shall be marked "FILL WITH DIESEL FUEL ONLY". Provisions shall be made for carrying the burner assembly on the drier-mixer. When the burner is connected to the fuel pump, the burner shall be equipped with a cut-off valve and oil resistance rubber fuel hose. The fuel hose shall be of sufficient length to reach all areas of the drier-mixer with the burner.

3.12 Bitumen metering system. The drier-mixer shall be furnished with a bitumen metering system. The metering system shall control the amount of bitumen being added to the mix from 3 to 12 percent by weight in increments of

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not more than 1 percent. The bitumen metering system shall consist essentially of a postive displacement pump, pump clutch, relief valve, a shutoff valve, plumbing hardware, and a volumetric flow-control system.

3.12.1 Bitumen pump and accessories. The drier-mixer shall be equipped with a positive-displacement pump. The engine power shall be transmitted to the pump through a friction clutch. The clutch shall be capable of transmitting maximum torque required to operate the pump when pumping asphalt cement having a penetration value between 85 and 100 heated to a temperature of not less than 225° F and at the highest volume specified herein. A positive lock or detent shall be provided to secure the clutch in both the engaged and disengaged position. The pump shall be equipped with a pressure relief valve, a shutoff valve, a flowmeter and a volumetric flow control. The pump pumping hardware shall include unions to permit removal of pump and accessories without complete disassembly of the plumbing.

3.12.2 Bitumen pump control valves. The bitumen pump shall be provided with ball type or lubricating plug type valves(s) installed to permit the following operations:

(a) Fill the bituminous tank from a supply tank.
(b) Circulate the bitumen being heated in the bituminous storage tank.
(c) Pump the required rate of bitumen for the mix being produced. The valve shall be opened or closed by one quarter turn of the valve control handle.

The direction of material flow through the valve ports shall be indicated by distinct markings on the valve control stem or control handle. An instruction plate made of the material specified in 3.21 shall be located near the valve control and the plate shall show the proper valve setting to perform the above specified operations. The plates shall be attached as specified herein. All controls for the pump shall be located so as to be accessible from the operator's station.

3.13 Aggregate charging hopper. The drier-mixer shall be equipped with an aggregate charging hopper. The charging hopper shall discharge directly into the drier or the combination drying and mixing chamber. When the charging hopper is used for measuring the amount of aggregate per mix, the charging hopper shall have a capacity of not less than 5 cubic feet and shall be clearly marked to indicate 3, 4, and 5 cubic foot levels. When used for only charging the drier, the charging hopper shall have a positive provision for continuously charging the aggregate into the drier to meet the rated capacity for the drier-mixer. Loading of the hopper shall be by hand shovel, directly from truck-bed level while the drier-mixer is coupled or uncoupled from the towing vehicle specified in 3.1.3.

3.14 Drives and bearings.

3.14.1 V-belt drives. V-belt drives shall be in accordance with MIL-B-52919 for drives using narrow V-belts (3V, 5V, and 8V cross section). All belt drives shall be matched multiple V-belt assemblies. Provisions shall be provided to

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adjust belt tension. Applying tension by use of an idler will be permitted providing the idler is used on the slack side of the belt. Sectional fiber link belting will not be permitted.

3.14.2 Chain drives. Chain drives shall be equipped with a tension adjustment for increasing or decreasing tension. Chains shall be of the roller-chain type conforming to ANSI B29.1. Sprockets shall have machine-cut teeth and the teeth shall be hardened.

3.14.3 Gear drives. Gears shall be enclosed in gear boxes and shall run in oil. Gears shall have machine-cut teeth and the teeth shall be hardened.

3.14.4 Shafts. All shafts shall be mounted on antifriction bearings.

3.14.5 Bearings. All antifriction bearings shall conform to FF-B-171, FF-B-185, or FF-B-187.

3.15 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each drier-mixer.

3.15.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.15.2 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.15.3 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipstick, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a hand held container.

3.15.4 Grease lubrication. All grease lubrication shall be with greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The drier-mixer shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

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3.15.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of military oil designated for use in the temperature range of 0°F. to 32°F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.15.6 Servicing and adjustments. The contractor shall service and adjust each drier-mixer for correct operation and travel including at least the following: Adjustment of the engine and of the drier-mixer machinery; adjustment of the brake system and electrical system; inflation of tires; and complete lubrication of the drier-mixer with the types and grades of lubricant recommended.

3.16 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.16.1.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Compostion II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102, Southern California Air Polution Control District.

3.17 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the drier-mixer shall conform to SAE J551, devices category.

3.18 Lifting attachments. Permanently affixed attachments that enable the drier-mixer to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.19.

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3.19 Tiedown attachments. Permanently affixed attachments that permit fastening drier-mixer to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.18.

3.20 Shipping data plates. Shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and shall show the silhouette of the drier-mixer in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The recommended lifting and tiedown arrangement including spreader bar arrangement shall be included on these plates. The data plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.21 Identification marking. The drier-mixer shall be identified in accordance with MIL-STD-130. The markings shall be applied to the drier-mixer on a plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.22 Control identification, safety and warning plates. A plate shall be furnished for each switch, gage, and control and at all points requiring attention for safe and efficient operation of the drier-mixer. The plates shall be labeled by lettering or other appropriate marking with information such as function, direction each control is to move, neutral position of each control, safety instructions, and other warnings or instructions, as necessary. The plates shall be anodized aluminum, sunken-etched or raised-relief. Plates that describe safety measures or that warn against operations detrimental to the drier-mixer shall be distinguished by the words "CAUTION" or "WARNING" in red lettering. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.23 Treatment and painting. The portions of the drier-mixer normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice, provided it is at least equivalent to MIL-T-704, Type A. Unless otherwise specified (see 6.2), the finishing coat shall be lusterless forest green in accordance with MIL-E-52929 Type II. Parts subject to temperatures of 400°F. or over shall be painted with heat resistant paint.

3.24 Stenciling. The gross weight of each drier-mixer shall be stenciled on each side of the drier-mixer. The prescribed tire pressure shall be stenciled on each fender. "LIFT" and "TIEDOWN", as applicable, shall be stenciled adjacent to their respective tiedown and lifting attachments. All stenciling shall be in block-type letters not less than 1 inch high, and shall be black in color.

3.25 Toolbox. A toolbox shall be provided and shall be of sufficient size to hold those tools and accessories necessary to operate the drier-mixer. The toolbox shall include provisions for drainage in its mounted position. The toolbox shall have a hinged lid with the hinges extending the full length of one edge and shall be provided with not less than two, tension-type, hook-latch

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fasteners that will keep the lid closed when the toolbox is subjected to vibration caused by secondary road travel. The toolbox lid shall have an overlapping lip to prevent entry of rain and dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of 90 degrees. The lid shall remain in the fully-open position by the weight of the lid or by detent brace holding devices.

3.26 Type I - Separate drier drum and pugmill.

3.26.1 Drier drum. The drier drum shall be a single, cylindrical drum suitable for heating and drying aggregate. The drum shall be fabricated by welding from steel plate. Lifting flights shall be provided within the drum to lift and drop the aggregate through the direct flame and hot gases of the burner. The drum shall be mounted on four chilled cast steel, or ductile iron trunnion rollers. Replaceable steel tires shall be provided around the shell of the drum for riding on the trunnion rollers. A discharge chute shall charge the pugmill or a batching hopper direct from the drier drum. A thermocouple shall be installed in the drier discharge chute with temperature registered on pyrometer in full view of the operator to provide for control of the final mix temperature. Stationary steel plates exposed to direct flame shall be insulated and shall be replacable.

3.26.2 Batching hopper. A batching hopper shall be provided when the charging hopper is not used for measuring the amount of aggregate per mix. The batching hopper shall have a positive method of measuring by weight or volume the amount of aggregate used per mix.

3.26.3 Pugmill. The pugmill shall be of the twin-shaft type with not less than four double-paddle arms or eight single-paddle arms on each shaft. Reversible and replaceable paddle tips shall be furnished on each paddle arm. The paddle tips shall be mounted at an angle with the axis of the pugmill shaft in order to produce a homogenous mixture with each aggregate particle throughly coated with bitumen. A jacket shall be provided around the pugmill to carry the hot burner exhaust gases and to form a heat-insulated pugmill. Pugmill liners shall be replaceable alloy steel plate not less than 3/16-inch thick. A discharge gate or positive method for regulating the discharge rate shall be provided. The discharge opening shall be located so that the mix can be directly discharged into areas to be patched and into wheelbarrows. The discharge opening shall be compatible with contractor's type wheelbarrows having tray sizes 34 to 41 inches long by 23 to 31 inches wide with the tray positioned either longitudinal or transverse with respect to the discharge opening. The discharge opening shall be located so that the wheelbarrow can be moved into and away from its loading position without twisting and manipulation and without interfering with personnel operating and discharge controls.

3.27 Type II - Combination drying and mixing chamber and pugmill.

3.27.1 Combination drying and mixing chamber. The chamber shall be constructed so that the drying of the aggregate and the mixing of aggregate with bitumen shall be accomplished in two separate cycles. The chamber shall be fabricated from steel plate. Chamber liners shall be replaceable alloy steel

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plate not less than 3/16-inch thick. The chamber shall have a gross volume of not less than 5 cubic feet. A discharge gate shall be provided which is capable of discharging all or a portion of each batch into a 3-cubic-foot wheelbarrow, having a tray size 34 to 41 inches long by 23 to 31 inches wide or directly into the area to be patched. The discharge opening shall be located so that a 3-cubic-foot wheelbarrow can be moved into and away from its loading position without twisting or manipulation. A thermocouple shall be installed in the drier discharge chute or drying and mixing chamber that will register the mix average temperature on a pyrometer located in full view of the operator.

3.27.2 Pugmill. The pugmill shall be an integral part of the combination drying and mixing chamber. The pugmill shall be of the shaft type with not less than eight paddle arms. A reversible and replaceable paddle tip shall be furnished on each paddle arm. The paddle tips shall be mounted at an angle with the axis of the pugmill shaft in order to produce a homogenous mixture with each aggregate particle throughly coated with bitumen.

3.28 Workmanship.

3.28.1 Steel fabrication. Steel used in the fabrication of the drier-mixer shall be free from kinks and sharp bends. Steel plate shall not be warped. The straightening of material shall be done by methods that will not cause injury to the metal. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.28.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washer or lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof of service loading.

3.28.5 Surfaces. All parts, components, and assemblies including castings, forgings molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

3.29 Transportability.

3.29.1 Rail hump test. When specified (see 6.2), the drier-mixer shall withstand a rail hump test as specified in 4.5.2.13, without permanent

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deformation to the drier-mixer. When commercial facilities are not available to conduct a rail hump test as specified, testing may be conducted at a Government facility.

3.29.2 Air transportability. The drier-mixer shall be air transportable in a C-130 aircraft of the dimensions shown in figure 3, except that a 6 inch clearance between the drier-mixer and the aircraft ceiling shall be maintained after loading.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection comparison (see 4.6).
(d) Inspection of packaging (see 4.7).

4.3 First article preproduction inspection.

4.3.1 Examination. The drier-mixer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The drier-mixer shall be tested as specified in 4.5.2.1 through 4.5.2.13. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each drier-mixer shall be examined as specified in 4.5.1 Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each drier-mixer shall be towed by the towing vehicle specified herein for a distance of not less than 10 miles over secondary roads and then operated for not less than 60 minutes without load. All controls shall be operated not less than 20 times each during the test to determine ease of operation, effectiveness, and responsiveness. During the roading test make at

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least four panic stops, and jackknife turns at angles up to 60 degrees from straight ahead both to the left and to the right. Turn the trailer lights on and check all circuits before the start of the test and repeat this procedure at the end of the test. Any damage, malfunction of the controls, brakes, lights, or excessive vibration shall constitute failure of this test.

4.5 Inspection procedure.

4.5.1 Examination. The drier-mixer shall be examined for the following defects:

101. Dimensions and weights not as specified.
102. Assembly not complete.
103. Material not as specified.
104. Safety devices missing or not as specified.
105. Ease of maintenance not as specified.
106. Trailer components not as specified.
107. Intervehicular connections not as specified.
108. Trailer electrical system not as specified.
109. Engine components not as specified.
110. Instrumentation not as specified.
111. Clutch not as specified.
112. Bitumen storage tank not as specified.
113. Burner heating system not as specified.
114. Fuel oil tank not as specified.
115. Bitumen metering system not as specified.
116. Control valves not as specified.
117. Aggregate charging hopper not as specified.
118. Drives and bearings not as specified.
119. Lubrication devices not accessible, missing, or not installed as specified.
120. Fungus and moisture resistance not as specified.
121. Tiedown and lifting attachments not as specified.
122. Identification markings missing or not as specified.
123. Transportation-data plates and operating instruction plate not as specified.
124. Treatment and painting not as specified.
125. Stenciling not as specified.
126. Toolbox missing or not as specified.
127. Type I or type II drier-mixer, as applicable, not as specified.
128. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The drier-mixer shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

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4.5.2.2 Road. The following road test shall be performed. Couple the trailer-mounted drier-mixer to the towing vehicle specified herein and tow the drier-mixer as specified in table IV, with the full complement of attachments, fuel, bitumen, and accessories stowed for travel.

TABLE IV. Road Test.

Course	Mileage and speed
Primary highway	200 miles at variable speeds up to 55 mph
Secondary road	100 miles at variable speeds up to 25 mph

During the road test, perform and record the actual results of the following:

(a) Measure and record the deviation or off tracking of the trailer with respect to the prime mover, when the trailer-mounted drier-mixer is towed straight ahead on a paved surfaced.
(b) Jacknife the trailer-mounted drier-mixer at angles up to 60 degrees from straight ahead both to the left and to the right.

4.5.2.2.1 Failure criteria. Failure to complete the test, inability to operate and function as specified herein, accessories damaged or not remaining in their carrying places shall constitute failure of this test.

4.5.2.3 Brake. Test the drier-mixer for conformance to the requirements specified herein:

(a) Tow the trailer-mounted drier-mixer over the test course and when the prime mover reaches a speed of not less than 20 mph, make a panic brake stop using both prime mover and trailer brakes. Perform the test four times, allowing not more than 3 minutes between tests.
(b) Tow the trailer-mounted drier-mixer at not less than 40 mph and apply service brakes to produce a deceleration rate of 10 feet per second, until the prime mover has slowed to 30 mph. Then immediately accelerate at wide-open-throttle again up to not less than 40 mph. Repeat the snubbing cycle for a total of 5 cycles and make a panic stop on the 6th cycle.

4.5.2.3.1 Failure criteria. Failure to complete the test, inability to function and operate as specified shall constitute failure of this test.

4.5.2.4 Bitumen storage tank capacity and heating test. The bitumen storage tank shall be filled with asphalt cement, grade 85 to 100 penetration, and the quantity required to fill the tank shall be determined. The contents of the tank shall be heated by means of the heating system provided and the time required to heat the contents from 200° F to 300° F shall be recorded. During the filling operations, place a 55 gallon drum of asphalt on the tank drum rack and empty contents into the tank. Fire the portable burner at maximum rate for a period of not less than 15 minutes. Determine the rate of fuel consumption.

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4.5.2.4.1 Failure criteria. Nonconformance to all requirements specified shall constitute failure of this test.

4.5.2.5 Bitumen metering test. Weight of bitumen based on final mix-batch weight from 3 to 12 percent by weight shall be determined and recorded in increments of 1 percent. Each of these predetermined weights of bitumen shall be metered by the metering pump, collected at the pugmill and weighed, and the weights shall be recorded.

4.5.2.5.1 Failure criteria. Failure of the metering device to supply the predetermined weights within plus or minus 5 percent shall constitute failure of this test.

4.5.2.6 Moisture content removal test. A mixture of sand and gravel conforming to the gradation requirement of table V, having an average surface moisture content of not less than 8 percent by weight shall be used for this test. Three tons of this mixture shall be processed through the complete cycle of the drier mixer in 1 hour or less. A complete cycle shall be defined as the summation of both the drying and mixing times. No bitumen shall be added to the aggregate during this test. The moisture content of the aggregate and the temperature of the aggregate at drier discharge or at the end of the drying cycle shall be determined not less than three times at equally spaced timed intervals throughout the test.

4.5.2.6.1 Failure criteria. Failure to process 3 tons of this material having an average residual surface moisture content of 1 percent or less in 1 hour or less, or temperature of the aggregate at drier discharge or at the end of the drying cycle not within the range of 275° F to 325° F shall constitute failure of this test.

4.5.2.7 Production capacity and performance test.

4.5.2.7.1 Production capacity. This test shall consist of processing 5 tons of aggregate, conforming to the gradation requirements of table V, through the drier-mixer into a hot bituminous mixture. Prior to drying, the aggregate shall have an average surface moisture content of 3 to 8 percent by weight. The binder shall be asphalt cement, grade 85 to 100 penetration, and shall be added to the aggregate in a proportion not greater than 8 percent or less than 7.5 percent by weight. The asphalt cement shall be added at a temperature within the range of 275° F to 325° F. The aggregate shall be loaded into the drier-mixer by hand shovel from truck bed level of the truck specified herein, and the hot bituminous mixture shall be discharged both into a contractor's type wheelbarrow and directly onto the ground. Prior to any hot bituminous mixture being produced, the residual surface moisture content and temperature of the aggregate at drier discharge or at end of the drying cycle shall be checked and, if necessary, drier or drying cycle shall be adjusted until residual surface moisture content is not greater than 1 percent and temperature of aggregate is within the range of 275° F to 325° F. The surface moisture content and temperature of the aggregate at drier discharge or at the end of the drying cycle shall be determined and recorded three additional times at equally spaced time intervals throughout the test. The actual time of drier-mixer operation shall be determined and recorded.

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4.5.2.7.1.1 Failure criteria. Nonconformance to 3.1.2 and 3.26 for type I drier-mixers or 3.1.2 and 3.27 for type II drier-mixers shall constitute failure of this test. Production of bituminous mixtures which are not homogeneous or which have aggregate particles not coated with bitumen as determined by visual inspection shall constitute failure of this test. The percentage of bitumen present in the bituminous mixture not as specified when checked by an extration test in accordance with AASHO Designation T58 shall constitute failure of this test.

TABLE V. Bituminous aggregate graduation requirements.

U.S. standard sieve and screens (square openings)	Percent passing by weight
3/4 inch	100
1/2 inch	80 - 100
3/8 inch	70 - 90
No. 4	50 - 70
No. 8	35 - 50
No. 30	18 - 29
No. 50	13 - 23
No. 100	8 - 16
No. 200	4 - 10

4.5.2.7.2 Performance. This test shall be an operational performance test for a period of not less than 40 hours. The test shall include cold mix production, road travel and dry mixing aggregate. The road travel time shall not be greater than 20 percent of the test hours specified herein. The aggregate dry mixing cycle time shall not be greater then 50 percent of the test hours specified herein and shall not be less than 5 cubic foot batch volume. The cold mix production shall be accomplished by processing aggregate, conforming to the gradation requirements of table V, through the drier-mixer into a cold mix bituminous mixture. The asphaltic material shall be added to the aggregate in a proportion not less than 5 percent nor greater than 6 percent by weight. The asphaltic material shall be added at a temperature within the range recommended by the Asphalt Instutute, Asphalt Plant Manual for the grade and type of asphaltic material used. The temperature of the aggregate at time of mixing shall be within plus or minus 25° F of the asphalt liquid temperature. The ability of the drier-mixer to produce cold mix at a rate of not less than 10 tons per hour shall be determined during this test.

4.5.2.7.2.1 Failure criteria. Failure to complete the test, inability to operate and function as specified, inability to produce cold mix at a rate of at least 10 tons per hour, malfunction of any control, or leakage of lubricants or asphalt liquid, shall constitute failure of this test.

4.5.2.8 Test procedures of lifting provisions. Test procedures of lifting provisions shall be as follows:

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(a) Lift the drier-mixer and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the drier-mixer. Determine the forward angles of application for each provision. Measure the clearance between each sling and the drier-mixer. Measure eye openings and clearance dimensions of the provision. Measure the load carried by each provision.

(b) Restrain the drier-mixer by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in paragraph (a) above. An alternate method of test is to lift the drier-mixer as in (a) above and add weights to the main frame until the required force is obtained. Hold the drier-mixer under either of the above loading conditions for at least 90 seconds.

4.5.2.8.1 Failure criteria. Weld failure of permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.9 Test procedures of tiedown provisions. With the drier-mixer anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown.

4.5.2.9.1 Failure criteria. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.10 Electromagnetic interference. Unless otherwise specified, the drier-mixer shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the preproduction model test report.

4.5.2.10.1 Failure criteria. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.11 Sound level measurement.

4.5.2.11.1 Exterior sound level. The contractor shall perform sound level tests in accordance with SAE J88. Certified prior test data is acceptable if available. Nonconformance to the requirements of 3.5.3.1 shall constitute failure of this test.

4.5.2.11.2 Personnel sound level. The drier-mixer shall be tested in accordance with MIL-STD-1474. Typical duty cycle tests shall not be used. Failure to meet the requirements of 3.5.3.2 constitutes failure of this test.

4.5.2.11.1 Failure criteria. Failure to meet the criteria specified in 3.5.3 shall constitute failure of the test.

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4.5.2.12 Aircraft clearance. Make a dimensional check of the drier-mixer and compare to the clearance diagram in figure 3.

4.5.2.12.1 Failure Criteria. Any dimension exceeding the clearance dimensions shown on figures 3 or nonconformance to 3.29 shall constitute failure of this test.

4.5.2.13 Rail hump test. When the rail hump test is specified (see 6.2) anchor the drier-mixer on a standard railway flat car. The loading, blocking and securing of the drier-mixer shall prevent any longitudinal, vertical, or lateral movement. Hump the flat car two times each at 8, 9 and 10 mph against two standing rail road cars with their brakes set in the emergency application position. The drier-mixer shall face in the direction of the two standing cars. The total weight of the two standing cars and loading shall be a miminum of two times but no greater than three times the weight of the flat car and test item. Repeat the procedure with the rear of the drier-mixer facing the two standing cars. The drier-mixer may be reversed by reversing it or by reversing the direction of the flatcar. Once the test has begun, there will be no readjustment of load, or any reconditioning of the blocks, chocking material or tiedowns except to reverse direction of the drier-mixer. After each impact, the drier-mixer and tiedowns shall be inspected to determine permanent deformation to components of the drier-mixer or other evidence of possible failure. After completion of this test, a 1-hour operational checkout will be performed to demonstrate proper operation of the drier-mixer and all controls.

4.5.2.13.1 Failure criteria. Weld failure or permanent deformation of the drier-mixer components or its tiedown attachments, or failure of the drier-mixer to operate after the test shall constitute failure of the test.

4.6 Inspection comparison. The Government may select drier-mixers at any time during the contract production period and subject these drier-mixers to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison drier-mixer to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison drier-mixer(s) and shall be cause for refusal by the Government to continue acceptance of production drier-mixers until evidence has been provided by the contractor that corrective action has been taken to eliminate the

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deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drier-mixers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all drier-mixers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Preproduction pack inspection.

4.7.1.1 Examination. Examine the preproduction pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completely processed drier-mixer prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

129.	Materials, methods or containers not as specified.	Each incorrect material, method, or container shall be considered one defect.
130.	Disassembly not as specified for level A or B.	
131.	Matchmarking tags not applied for level A or B.	
132.	Matchmarking information not applied with a waterproof ink for level A or B.	
133.	Unprotected exterior metal surfaces not coated with preservative as specified for level A or B.	
134.	Preserved machined surfaces not wrapped or covered with barrier material as specified for level A or B.	
135.	Trailer chassis components not preserved in accordance with the referenced document for level A or B.	
136.	Engine and engine components not preserved in accordance with the referenced document for level A or B.	
137.	Gear cases not filled to operating level and tagged as specified for level A or B.	
138.	Gears and chains enclosed within housing not actuated to assure coverage for level A or B.	
139.	Exposed chains not preserved as specified for level A or B.	
140.	Drive belts not removed or released from tension for level A or B.	
141.	Burner heating system not preserved as specified for level A or B.	

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142. Exhaust stack opening not covered as specified for level A or B.
143. Maintenance tools not preserved in accordance with the referenced document for level A or B.
144. Components requiring protection and not specifically identified, not preserved as specified for similar components for level A or B.
145. Consolidation not as specified for level A or B.
146. Secured components interfere with lifting or towing for level A or B.
147. Strapping not finish B for level A.
148. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The preproduction pack may be accomplished utilizing either the preproduction drier-mixer or the production drier-mixer. If the preproduction model drier-mixer is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model drier-mixer and the production drier-mixer.

5.2 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The drier-mixer shall be disassembled to meet the transportability requirements specified in 3.29. Any additional disassembly shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage, and loss and protruding parts, such as the exhaust stack, that increase cubage and that can be removed and reassembled without special tools or skills. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent loss. Removed keys shall be secured to the keyway of the primary component or shaft.

5.2.1.2 Matchmarking. Parts removed and mating parts on the drier-mixer shall be matchmarked when necessary to facilitate reassembly. The parts shall be identified by tags conforming to UU-T-81, type A, and the tags attached in a manner to prevent damage to the preservation or to the parts. The printing on the tags shall be applied with waterproof ink.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

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5.2.1.4.1 Unfinished surfaces. Unpainted or unfinished exterior metal surfaces such as adjusting rods, tie rods, hand cranks, trunnion rollers, blower, nuts, bolts, pins, linkages, and any other similar parts or surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

5.2.1.4.2 Machined surfaces. Exposed machined metal surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, grade A, type I, class 2 and the material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.5 Trailer chassis. The components of the trailer chassis shall be preserved in accordance with the leval A (mobile preservation and packaging requirements of MIL-STD-281.

5.2.1.6 Engine. The engine and engine accesories shall be preserved in accordance with level A requirements of MIL-E-10062, as specified for type I, method I. Except when a liquid cooled engine is provided, the engine cooling system shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153.

5.2.1.7 Gears.

5.2.1.7.1 Enclosed gears. The gear housing shall be filled to the operating level with the lubricant recommended for operation and the gears actuated to assure coverage of all interior surfaces with the lubricant. A cloth tag conforming to UU-T-81, shall be attached in a conspicuous location indicating "The housing is filled to the operating level with the required operational lubricant." The printing on the tag shall be applied with waterproof ink.

5.2.1.7.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.8 Drive chains.

5.2.1.8.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushing. The excess preservative shall be drained, and the entire chain shall be coated with type P-1 preservative. Removed chains shall be coiled, and the coil shall be secured with soft annealed wire and wrapped with barrier material conforming to MIL-B-121, type 1, grade A, class 2, secured with tape conforming to PPP-T-60, type IV. Sprockets shall be coated with type P-1 preservative.

5.2.1.8.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.2.1.7.1 for enclosed gears and gear housings.

5.2.1.9 Shafts and universal joints. Shafts and universal joints shall be coated with type P-1 preservative.

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5.2.1.10 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. Pulley grooves shall be coated with primer conforming to TT-P-664. The belts shall be tagged or identified to indicate their reinstallation positions.

5.2.1.11 Burner heating system. Interior surfaces of the pumps, lines, and burner shall be coated with lubricating oil conforming to MIL-L-21260, type I, grade 30.

5.2.1.11.1 Exhaust stack. The exhaust stack, placed in the travel position shall have the opening covered with barrier material conforming to PPP-B-1055, class E-2. The barrier material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.12 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.13 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.2.1.14 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.15 Miscellaneous preservation. Components not specifically identified herein requiring protection from corrosion and mechanical damage shall be preserved as specified for components of similar design or construction.

5.2.1.16 Consolidation. Loose components, including those removed during disassembly shall be consolidated together in the toolbox when space permits. When space does not permit, the remainder of the items that will not fit in the toolbox shall be consolidated together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, grade B; or secured to the drier-mixer in a protected location. Securing of the items or boxes to the drier-mixer shall be with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, in a manner that will not increase cubage or interfere with lifing or towing the drier-mixer. The toolbox lid shall be secured to prevent pilferage or accidental opening.

5.2.2 Level B. Each complete drier-mixer shall be preserved as specified in 5.2.1 for level A, with the following exceptions.

5.2.2.1 Trailer chassis. The components of the trailer chassis shall be preserved in accordance with the level B (mobile) preservation and packaging requirements of MIL-STD-281.

5.2.2.2 Exposed chains. Exposed drive chains shall be coated with type P-3 or P-9 preservative only.

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5.2.2.3 Drive belts. Removed belts shall not require preservation but shall be either consolidated or secured to the drier-mixer in a protected location near the pulley for which intended.

5.2.2.4 Consolidation. Boxes shall be domestic type and strapping for securing need not be finish B.

5.2.3 Level C. Each complete drier-mixer and all items furnished with the drier-mixer shall be preserved to afford protection against mechanical and physical damage. Items with critical surfaces required for fit or functions shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping or cartonizing of the individual item. Liquid cooling systems shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before damaged. All lubrication reservoirs sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used, the temperature range of lubricant used and grade of lubricant used. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and any other servicing required prior to placing the drier-mixer in operation.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each complete drier-mixer, preserved as specified in 5.2, shall be prepared for unboxed (mobile) shipment in accordance with the applicable rules and regulations of the carrier as required for overseas shipment.

5.3.2 Level C. Each complete drier-mixer preserved as specified in 5.2, shall be prepared for unboxed (mobile) shipment in accordance with Uniform Freight Classification rules or National Motor Freight Classification rules as applicable.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The drier-mixer is intended to produce bituminous surfacing material for use in the maintenance and repair of road and airfield pavements and is roadable for travel between operating sites.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) Type drier-mixer required (see 1.2).

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(c) Date of issue of DoDiss applicable to this contract and exceptions thereto (see 2.1.1).
(d) When preproduction model is not required (see 3.2).
(e) Time frame for submission of production model and number of drier-mixer required (see 3.2).
(f) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(g) When the conversion to the 24 volt electrical system is not required (see 3.7.14.1).
(h) Type of engine required (see 3.8).
(i) When storage batteries shall be other than specified (see 3.8.3).
(j) When fungus and moisture resistance is not required (see 3.16).
(k) When electromagnetic interference requirements are other than as specified in 3.17.
(l) When the finishing coat shall be other than lusterless forest green (see 3.23).
(m) When rail hump testing is required. If commercial facilities are not available, what Government facilities will be used (see 3.29.1).
(n) Time frame required for submission of the preproduction pack (see 5.1).
(o) Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 Government loaned property. The contracting officer should arrange to loan to the contractor a dump truck as specified in 3.1.3.

6.4 Preproduction model. Any changes or deviations of production drier-mixers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish drier-mixers conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manual to be furnished with each drier-mixer.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of drier-mixers.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the drier-mixers in accordance with this specification.

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6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME
Air Force - 99

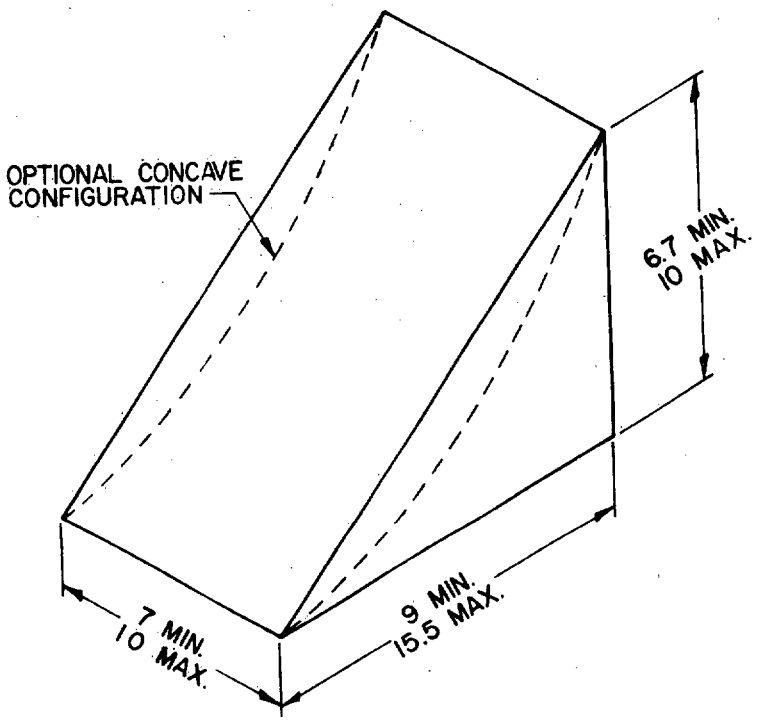
Review activity:
Army - AT

User activity:
Navy - YD

Preparing activity:
Army - ME

Project 3895-0256

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT-11.5 LB.

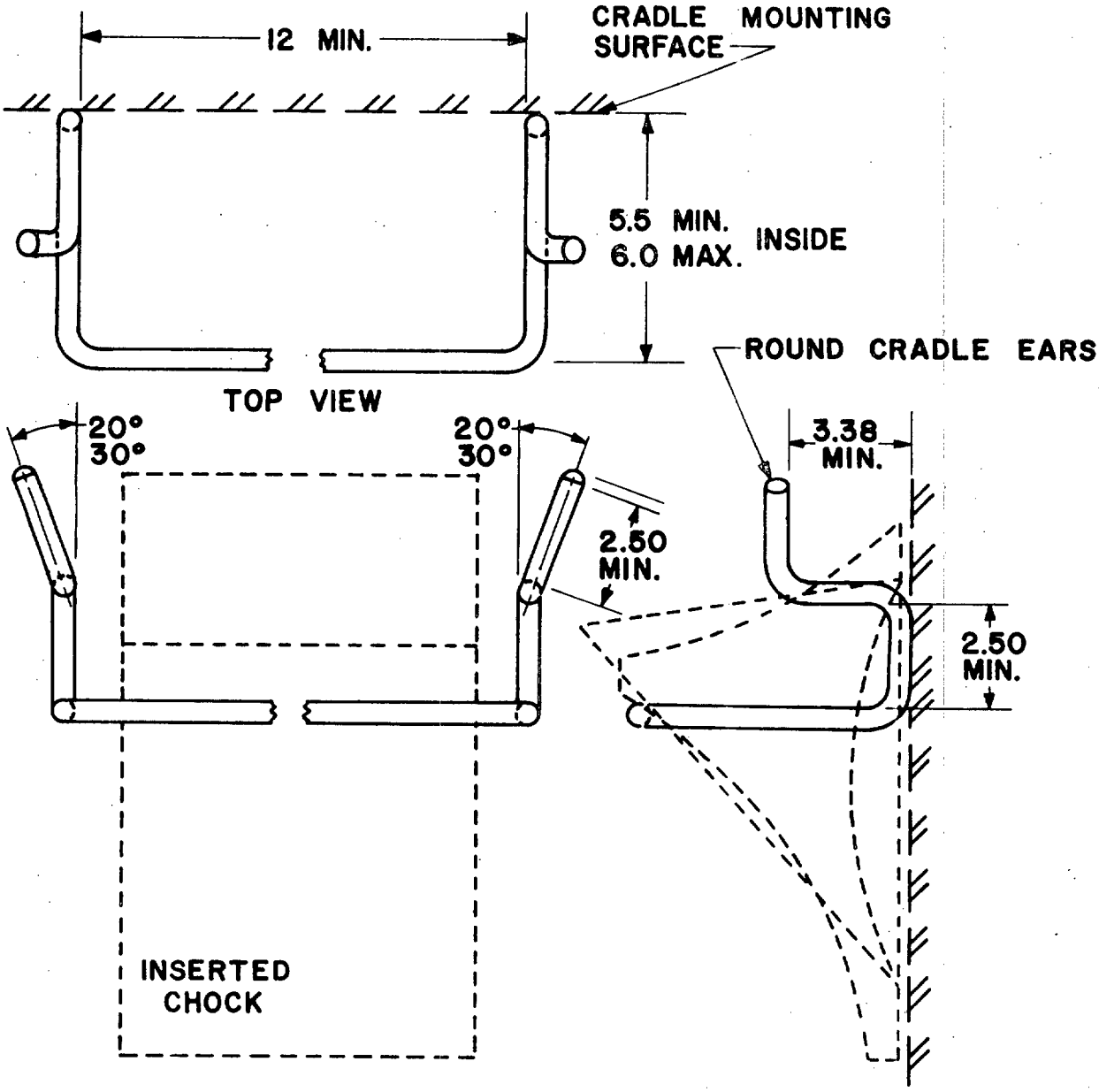
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 1. Chock block assembly.

X 603 A

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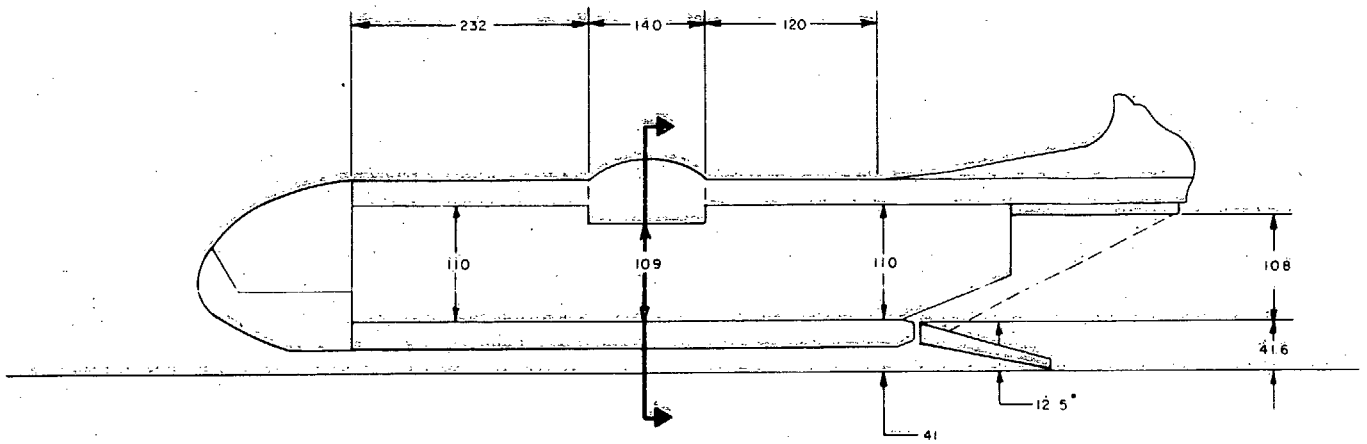


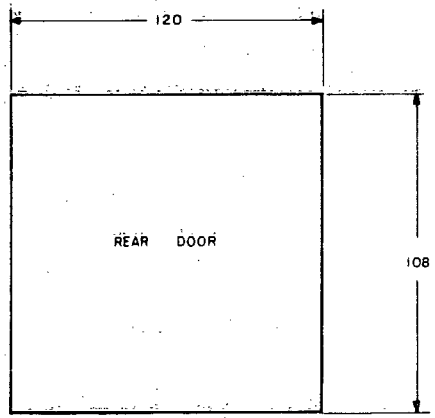
CRADLE MATERIAL NOTE: MINIMUM 1/2-INCH DIA. STEEL ROD. BEND RADII-1.0-IN. MIN.

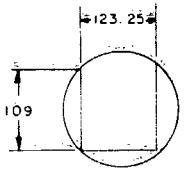
FIGURE 2. Cradle.

X-604A

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SECTION A A

NOTE: UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES.

FIGURE 3. Cargo compartment profile (C-130).

X-195A

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